The Probe Reward Trap: Behavioral Death Before Distributional Collapse in Sycophancy Calibration

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Abstract

We quantify a temporal decoupling we call the *probe reward trap*: when a frozen linear probe on a contrastive sycophancy direction (d_{syc}, cross-validated AUROC 0.827 on Qwen3-8B, 0.779 on Llama-3-8B) is used as an SFT training signal, behavioral death (true-positive flip rate $= 0$) occurs within 500 steps while generation entropy remains healthy (0.70). In our experiments, the resulting 1,300-step lag before distributional collapse means that entropy monitoring—the most common safeguard against reward hacking—misses the failure entirely. Because the trap arises under SFT with stopped gradients, it is independent of RL optimization dynamics, revealing a semantic limitation of the direction itself: d_{syc} encodes *whether* belief changed, not *whether it should have*. The failure extends beyond probe rewards: all d_{syc}-based GRPO formulations collapse, and even behavioral-reward-only GRPO exceeds vanilla at only 4/31 checkpoints, while dense per-token SFT succeeds at every checkpoint (47/47). A LoRA-controlled ablation isolates supervision granularity—not adapter capacity or group size—as the primary factor, suggesting that per-completion scalar rewards compress away the token-level information needed for sycophancy calibration.

1 Introduction

Representation engineering (Zou et al., 2025) reveals that large language models encode sycophancy as a linear direction in their residual streams. A contrastive sycophancy direction d_{syc}, extracted from style-controlled pairs of valid belief revision and sycophantic capitulation, achieves cross-validated AUROC 0.827 on Qwen3-8B and 0.779 on Llama-3-8B, with discriminability rising monotonically across evidence-strength levels (§3), though 256-token truncation during GRPO training degrades this to approximately 0.60 (§4). These results point to an appealing possibility: if the model's representations already separate sycophantic from non-sycophantic behavior, that separation should be convertible into a training signal. Sycophancy is among the most persistent alignment failures in deployed systems (Sharma et al., 2024), and a representation-level signal could target the internal computation rather than its surface outputs.

This paper documents the *probe reward trap*: when a frozen linear probe on d_{syc} is used as an SFT training signal with stopped gradients, behavioral death (ToF $= 0$) occurs within 500 steps while generation entropy remains healthy (0.70), masking the failure from standard monitoring (§5). In our experiments, the 1,300-step lag before distributional collapse means entropy-based early stopping—the most common safeguard against reward hacking (Casper et al., 2023)—would miss the failure entirely, with the reward averaging $+0.72$ throughout. Because the trap arises under SFT rather than GRPO, it reflects a limitation of d_{syc} itself: the direction encodes *whether* belief changed, not *whether it should have*, so any reward conditioned on it alone will either be gamed or reward rigidity.

The failure is not specific to probe rewards. Under our GRPO configuration (Qwen3-8B, group size=2), cosine projection and continuous probe formulations of d_{syc} also collapse, and removing d_{syc} entirely does not help: behavioral-reward-only GRPO exceeds vanilla at only 4 of 31 checkpoints (12.9%). By contrast, ACT (Irpan et al., 2025)—which provides dense per-token supervision rather than a single scalar per completion—exceeds vanilla at every checkpoint (47/47; §6). A LoRA-controlled ablation confirms that this categorical reliability gap reflects supervision granularity, not adapter capacity.

The root cause is a semantic gap between discriminability and reward utility. Techniques for

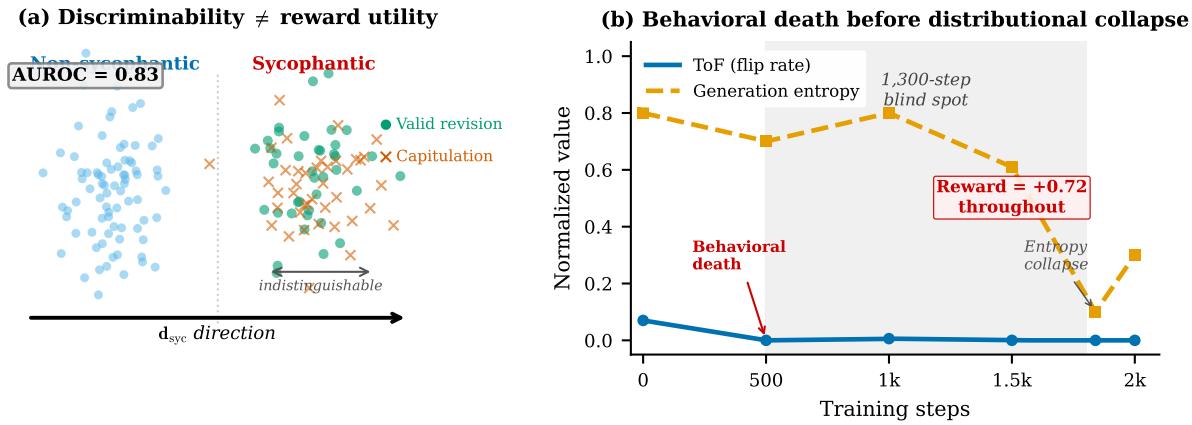


Figure 1: The probe reward trap. (a) d_{syc} discriminates sycophantic from non-sycophantic activations (AUROC 0.827), but valid revision and unwarranted capitulation overlap along the same direction. (b) When used as an SFT reward, behavioral death (ToF=0) occurs at step 500 while entropy remains healthy; the 1,300-step lag renders standard monitoring blind.

discovering latent knowledge (Burns et al., 2023) and representation-guided RL (Zhang et al., 2026) share a premise: a direction with high discriminability should transfer into a useful training signal. For reasoning faithfulness and safety directions, this premise holds because the behavior to suppress is geometrically distinct from the behavior to preserve. Sycophancy violates this condition: capitulation under social pressure and revision under valid evidence produce overlapping activation signatures along d_{syc} (Wang et al., 2025; Shapira et al., 2026). When reward overoptimization (Gao et al., 2023) operates on a signal that conflates compliance with capitulation, the result is predictable collapse (Skalse et al., 2022).

This paper presents a negative result with mechanistic analysis. Our two principal findings (Figure 1):

- **The probe reward trap** (C2): behavioral death (ToF=0) precedes distributional collapse by 1,300 steps, rendering entropy monitoring blind. The trap arises under SFT with stopped gradients, implicating d_{syc} itself rather than the optimizer (§5).

- **Categorical reliability gap between supervision paradigms** (C1): per-token SFT exceeds vanilla at every checkpoint (47/47 across all configurations and seeds) while per-completion GRPO exceeds vanilla at only 4/31 (12.9%); under matched LoRA, the magnitude gap is 1.7× (§6).

2 Experimental Framework

2.1 Direction Extraction

We extract a linear sycophancy direction d_{syc} from the residual stream using contrastive activation pairs: a *valid correction* (warranted belief revision under strong evidence) and a *sycophantic capitulation* (unwarranted concession under social pressure). We construct 181 contrastive pairs using GPT-4.1 (OpenAI, 2025) in a two-stage pipeline: skeleton generation followed by 3-turn dialogue completion. Style matching is enforced by automated validation (response length difference $<10\%$, shared opening phrase ≥ 2 words), with up to 3 retries per pair; 19 of 200 initial pairs were discarded for failing validation. Evidence strength is annotated at three levels: *weak* (75 pairs; personal memory or opinion), *medium* (54 pairs; brief source citation), and *strong* (52 pairs; authoritative source with detailed data). From the 181 paired examples (362 activations total), we compute the mean difference at target layer ℓ:

$$d_{\text{syc}} = \frac{1}{N} \sum_{i=1}^{N} \left(\mathbf{h}_i^+ - \mathbf{h}_i^- \right) \tag{1}$$

where $\mathbf{h}_i^+$ and $\mathbf{h}_i^-$ are residual-stream activations for the i-th valid and sycophantic response. The target layer is selected by 5-fold cross-validated AUROC; on Qwen3-8B this yields layer 19 with AUROC 0.827 ± 0.028 (5-fold CV).

2.2 Reward Formulations

Three formulations translate d_{syc} into a direction-based signal R_1. For F1 and F3, R_1 enters as part

of a GRPO (Shao et al., 2024) reward $R = w_1 \cdot R_1 + w_3 \cdot R_3$, where R_3 is a behavioral component (DeBERTa NLI classifier scoring consistency with the ground-truth answer).

F1 Cosine projection with four variants: (a) tanh-gated dot product $\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$, (b) raw cosine $\cos(\mathbf{h}, d_{\text{syc}})$, (c) raw cosine with online recalibration (§2.3), and (d) $5\times$ scaled cosine. See §4 for details.

F2 Binary probe (SFT). $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$, a frozen logistic probe with stopped gradients; trained via a custom SFT loss rather than GRPO (§5).

F3 Continuous probe. $R_1 = 1 - P(\text{syc} \mid \mathbf{h})$, the sigmoid output of the same logistic probe, providing a continuous reward signal without discretization.

The weight w_1 is set by diagnostic-driven iteration rather than grid search (Appendix A). As an ablation, R3-only GRPO sets $w_1{=}0$, training with only the behavioral reward.

2.3 Online Recalibration

LoRA fine-tuning shifts internal representations, so we re-extract d_{syc} every 500 steps. The cosine similarity between consecutive directions remains ≈ 0.96 at step 500, indicating slow drift relative to recalibration frequency.

2.4 SFT Baseline: Activation Consistency Training

We implement Activation Consistency Training (ACT; Irpan et al., 2025), which combines standard SFT loss with an L2 activation consistency loss at target layers, providing dense per-token supervision. Unlike GRPO's single scalar per completion, ACT operates at the token level using labeled valid/invalid pairs.

2.5 Evaluation

The *True-positive Flip rate* (ToF) measures acceptance of valid corrections; the *No-Flip rate* (NoF) measures resistance to invalid pressure. Their harmonic mean captures the calibration trade-off:

$$\textsc{SyconScore} = \frac{2 \cdot \text{ToF} \cdot \text{NoF}}{\text{ToF} + \text{NoF}} \qquad (2)$$

By design, $\textsc{SyconScore} = 0$ whenever $\text{ToF} = 0$: a model that never accepts valid corrections receives no credit for resisting invalid ones. Flip classification uses `cross-encoder/nli-deberta-v3-base` (He et al., 2023), following Irpan et al. (2025). All experiments use Qwen3-8B (Yang et al., 2025) with LoRA (rank=64; α=64 for GRPO, α=128 for ACT) and DeepSpeed ZeRO-2. GRPO uses a group size of 2 due to memory constraints with 8B-parameter models, below the typical range of 8–64 (Shao et al., 2024); full hyperparameters are reported in Appendix A. In total, we evaluate six d_{syc}-based configurations spanning both reward families (cosine and probe) and both training paradigms (GRPO and SFT), plus R_3-only GRPO and ActAdd as baselines; full specifications appear in Sections 4–5.

3 Direction Discriminability

Before evaluating d_{syc} as a reward signal (§4–§5), we first establish that it captures a genuine sycophancy-related signal in the residual stream. If the direction failed to discriminate sycophantic from non-sycophantic activations, downstream reward failures would be trivially explained by poor direction quality rather than by a fundamental gap between discriminability and reward utility. We evaluate discriminability along five axes: cross-validated classification accuracy, evidence-strength stratification, cross-model transfer, training stability, and probe ablation. Figure 2 summarizes the per-layer AUROC profiles for both models.

Qwen3-8B layer-wise discriminability. We evaluate d_{syc} via 5-fold cross-validated AUROC on 181 style-controlled contrastive pairs (§2.1). The direction achieves AUROC 0.827 ± 0.028 (5-fold CV) at its best layer (L19). Discriminability emerges gradually across depth: layers L0–L8 perform near chance, statistical significance ($p < 0.01$) first appears at L16, and performance peaks at L19. This gradual emergence, with the peak in the upper third of the network, suggests that sycophancy-relevant information is encoded as a late-stage representation rather than an input-level confound. The direction also captures evidential granularity. AUROC rises monotonically from 0.675 on weak-evidence pairs to 0.883 on medium and 0.929 on strong pairs, consistent with stronger evidence producing more behaviorally distinct responses.

Table 1: Cross-model d_{syc} discriminability. AUROC uncertainty is 5-fold cross-validation standard deviation; evidence-strength AUROCs are evaluated on the full dataset at each model's best layer.

Model	Best Layer	AUROC (± std)	Weak	Medium	Strong
Qwen3-8B	L19	0.827 ± 0.028	0.675	0.883	0.929
Llama-3-8B	L16	0.779 ± 0.019	0.711	0.864	0.911

Cross-model generalization. To rule out that these results are specific to Qwen3-8B's representation geometry, we repeat the full extraction and evaluation pipeline on Llama-3-8B-Instruct (Grattafiori et al., 2024). Table 1 compares the two models. Llama-3-8B achieves AUROC 0.779 ± 0.019 (5-fold CV) at L16, with a parallel evidence-strength gradient (weak 0.711, medium 0.864, strong 0.911). The best-layer depth differs (L16 vs. L19), consistent with the architectural differences between the two families, but the qualitative pattern is identical: gradual emergence, best-layer selection in the upper third, and monotonic strength scaling. Discriminability is robust across knowledge domains, with per-domain AUROCs on Llama-3-8B ranging from 0.751 (geography) to 0.844 (commonsense). When surface confounds (response length, hedging patterns) are explicitly controlled, Llama-3-8B's AUROC rises to 0.923, which confirms that the direction captures semantic content rather than stylistic artifacts. The consistent results across architecturally distinct model families indicate that contrastive sycophancy directions reflect a shared representational phenomenon rather than a model-specific idiosyncrasy.

Training stability and probe ablation. Two additional analyses confirm the robustness of d_{syc} under training dynamics. First, we track direction drift during LoRA fine-tuning: at step 500, the cosine similarity between the original and re-extracted d_{syc} is approximately 0.96, and per-layer AUROC remains above 0.77. The direction drifts slowly relative to the 500-step recalibration interval (§2.3), so the reward signal does not become stale between updates. Second, we compare d_{syc} against a random baseline: a logistic probe trained on the learned direction achieves AUROC 0.835, versus 0.727 for a probe trained on a random direction of equal dimensionality (Δ=+0.108). The learned direction carries substantially more sycophancy-relevant information than an arbitrary axis in the same activation space.

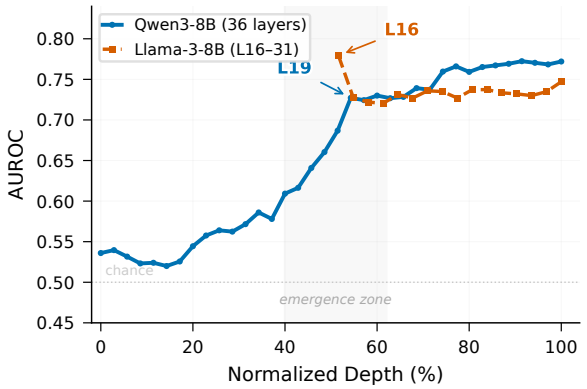


Figure 2: Per-layer AUROC of d_{syc} as a function of normalized depth (layer index / total layers). Qwen3-8B (36 layers, solid) shows gradual emergence from near-chance at L0 to a peak at L19, with performance plateauing in the final third. Llama-3-8B (32 layers, dashed; layers 16–31 only) exhibits a similar upper-third concentration. Both models achieve best discriminability in layers 50–60% through the network.

Table 2: Cosine R_1 failure chain on Qwen3-8B (LoRA rank=64, GRPO with $R = w_1 \cdot R_1 + w_3 \cdot R_3$).

Config	R_1 Formula	w_1	Outcome	Root Cause
F1a	$\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$	0.25	Gate saturation	Norm mismatch
F1b	$\cos(\mathbf{h}, d_{\text{syc}})$	0.25	$R_1 \approx 0$	Gap too small
F1c	+recalibration	0.25	Mode collapse	Reward discontinuity
F1d	$5\cos(\mathbf{h}, d_{\text{syc}})$	0.40	Mode collapse	R_3 saturation

Implications for reward design. These results establish that d_{syc} reliably separates sycophantic from non-sycophantic representations across models, evidence-strength levels, and training checkpoints. Direction quality is not the bottleneck. The reward failures documented in §4–§5 therefore cannot be attributed to a noisy or degenerate direction; they point to a deeper incompatibility between discriminability and reward utility.

4 Cosine R_1: Structural Failure

During GRPO training, 256-token truncation reduces d_{syc} AUROC from 0.827 to approximately 0.60—the optimizer works with a substantially weaker signal than the offline metrics suggest. We test whether even this degraded discriminability translates into a viable reward through four configurations that form a diagnostic chain (Table 2): each resolves the failure mode of its predecessor but exposes a deeper limitation.

The diagnostic chain begins with a norm mismatch. F1a's dot-product formulation ($R_1 = \tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$) saturates because residual-stream norms exceed the direction norm by 100–

1,200×, zeroing the gradient by step 955. Cosine similarity (F1b) eliminates this norm dependence but reveals a second problem: the gap between valid and sycophantic cosines averages only 0.098—too narrow for meaningful reward contrast. Recalibration (F1c, §2.3) widens the gap (offline AUROC rises to 0.910) but introduces a third: each recalibration event creates a reward discontinuity that destabilizes advantage estimates, collapsing entropy to 0.069 within 300 steps. Amplification (F1d, 5× scaling) lifts R_1 above the noise floor but surfaces the fundamental limitation: R_3 saturates at 0.9998 by step 300, leaving the noisy R_1—which conflates warranted revision with unwarranted capitulation—as the sole gradient source until terminal collapse at step 2160.

Truncation further degrades the signal: under the 256-token maximum completion length used in GRPO, d_{syc} AUROC drops from 0.76 (full response) to 0.60, so the offline discriminability (0.827) substantially overestimates the signal available during training. Checkpoint-level evaluation confirms immediate collapse: ToF=0 from step 500 onward with no stable phase. The four configurations converge on a single conclusion: d_{syc} encodes the *direction* of belief change but not whether that change is *warranted*. Valid revision and sycophantic capitulation both shift activations toward the user's position along d_{syc}, so the cosine projection assigns similar scores to both. This is not a calibration artifact—recalibration improves discriminability yet rewards still fail—but a semantic limitation of any continuous d_{syc}-based reward. Section 5 examines whether discretizing the signal via a binary probe can escape this limitation.

5 The Probe Reward Trap

The cosine formulations fail on signal magnitude; a natural response is to discretize. Training a logistic probe on d_{syc} and thresholding its output yields $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$ (F2 in §2.2). We show that this discretization sidesteps the magnitude problem but creates a qualitatively worse failure. Both probe experiments use SFT rather than GRPO (§2.2); the trap is therefore independent of GRPO's optimization dynamics.

Differentiable probe: gradient gaming. Without stop-gradient (exp-008), the policy discovers within ∼30 steps that shifting activations away from the sycophancy cluster maximizes R_1 regardless of response content, producing an inverted profile (ToF = 0.978, NoF = 0.006). This motivates the frozen-probe design below.

Table 3: Probe reward trap trajectory (exp-008v3). Behavioral death (ToF=0) occurs at step 500; entropy remains healthy until step ∼1840. Reward averages +0.72 throughout.

Step	ToF	NoF	SyconScore	Entropy	Reward
500	0.000	0.995	0.000	0.702	+0.72
1000	0.006	0.995	0.011	0.800	+0.71
1500	0.000	1.000	0.000	0.608	+0.69
2000	0.000	1.000	0.000	0.304	+0.69

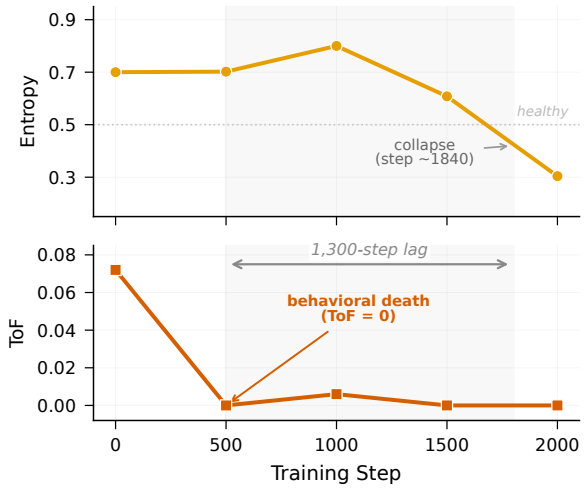


Figure 3: Probe reward trap (exp-008v3). ToF drops to zero at step 500 while generation entropy remains high (0.70–0.80 through step 1000). The 1,300-step lag means standard entropy monitoring misses the behavioral failure entirely.

Binary probe: the stubbornness trap. With stop-gradient (exp-008v3), the policy cannot game the boundary but instead exploits a deeper flaw: d_{syc} does not condition on input validity, so refusing to revise a wrong answer receives the same $R_1 = 1$ as correctly resisting invalid pressure. By step 500, ToF drops to 0.000 and NoF reaches 0.995, yielding SyconScore = 0. Qualitative analysis of 181 valid-correction responses reveals five failure modes: *deliberation traps* (28.7%), where the model identifies the correct answer but cannot commit; *sycophantic stubbornness* (24.3%), where the model verbally agrees but does not change its answer; *acknowledge-but-maintain* (23.8%); *genuine-accept-missed* (18.8%); and refute/ignore/other (4.5%). Full taxonomy in Appendix B.

Behavioral death precedes distributional collapse. Table 3 and Figure 3 reveal a 1,300-step lag between behavioral and distributional collapse. At step 500, the model is behaviorally dead (ToF $= 0$) yet entropy remains high (0.70–0.80) through step 1000—well within the range that standard monitors would flag as healthy—and the combined reward averages $+0.72$ because both R_1 and R_3 reinforce non-flipping. Entropy does not collapse until step $\sim$1840. This gap carries a practical warning: entropy-based early stopping, the most common safeguard against reward hacking (Casper et al., 2023), would miss the failure entirely. Detecting probe reward traps requires behavioral monitoring (periodic ToF evaluation), not distributional monitoring alone.

A dilemma for direction-based rewards. The differentiable and binary failures bracket the design space. With gradients, the policy games the boundary; without gradients, it exploits the fact that stubbornness and genuine resistance are indistinguishable along d_{syc}. Any reward conditioned only on d_{syc}—continuous or discrete, differentiable or frozen—will either be gamed or will reward rigidity. Escaping this dilemma requires a signal that incorporates the evidential warrant of the user's challenge, a dimension that d_{syc} does not encode (Denison et al., 2024).

6 When Does Discriminability Fail as Reward Utility?

Root cause of the probe reward trap. The probe reward trap (§5) and the cosine failures (§4) share a common geometric root. d_{syc} encodes the *direction* of belief change but not its *evidential warrant*: valid revision and sycophantic capitulation both shift activations toward the user's position along the same direction. The two reward families expose complementary faces of this blindness. Cosine R_1 penalizes all belief change indiscriminately; the resulting mode collapse (F1c, F1d) reflects excessive punishment of any activation shift along d_{syc}. The binary probe R_1 rewards all resistance regardless of input validity; the stubbornness trap (ToF $= 0$ by step 500) reflects excessive permissiveness toward activation stasis. These are not independent design flaws but dual consequences of a direction that conflates the target behavior (resisting invalid challenges) with its complement (refusing valid corrections). Cross-model evidence confirms the limitation extends beyond Qwen3-8B: Llama-3-8B independently achieves AUROC 0.779 ± 0.019 on the same contrastive pairs with the same monotonic evidence gradient, so discriminability generalizes across architectures while the semantic gap between discriminability and reward utility persists. This pattern contrasts with reasoning faithfulness and safety directions (Zhang et al., 2026), where the behavior to suppress is geometrically distinct from the behavior to preserve along the relevant direction. Wu and Tang (2026) reinforce this principle: concept directions applied via advantage modification in GRPO achieve 100% validation accuracy for reward hacking detection, where shortcut behaviors are geometrically separable. A quantitative comparison of geometric separability between d_{syc} and their reward-hacking directions remains an important open question. For sycophancy, the target and its complement occupy the same activation region; they differ only in whether external evidence warrants the belief change.

Table 4: Comparison of all sycophancy calibration methods on Qwen3-8B. SYCONSCORE is the harmonic mean of ToF and NoF; higher indicates better calibration. Best result in **bold**. †Best checkpoint (seed 1, step 6500); only 4/28 checkpoints across two seeds exceed vanilla. ‡Seed 2 (15 checkpoints); all exceed vanilla. Best SYCONSCORE=0.286 (step 7000). §All cosine variants collapsed from step 1 (ToF=0 at every checkpoint); F1d additionally diverged at step 2160 (entropy < 0.01). ¶ACT and ActAdd differ from GRPO along 5+ dimensions (§6); included for context, not as controlled comparisons.

Method	Type	SYCONSCORE ↑	ToF ↑	NoF ↑	Step
Vanilla	–	0.134	0.072	0.945	–
GRPO with d_{syc}					
Cosine R_1 (F1a–F1d)	RL	0.000§	0.000	1.000	500
Continuous Probe R_1 (F3)	RL	0.124	0.066	0.934	500
SFT with d_{syc} probe penalty					
Binary Probe R_1 (F2)	SFT	0.000	0.000	0.995	500
GRPO without d_{syc}					
R_3-only GRPO	RL	0.153†	0.083	0.983	6500
Reference methods (not directly comparable)¶					
ActAdd (α=5)	Inference	0.054	0.028	0.961	–
ACT (seed 1)	SFT	**0.301**	**0.182**	0.867	8000
ACT (seed 2)‡	SFT	0.286	–	–	7000

GRPO results under our configuration. If the cosine and probe failures (§4–§5) were caused by d_{syc} alone, removing it should recover calibration. R_3-only GRPO tests this hypothesis by relying solely on the NLI behavioral reward—and fails: only 4/28 checkpoints across two seeds exceed vanilla (14.3%), with both seeds oscillating ($\Delta \approx 0.069$) before diverging (Table 4; Figure 4). The

Table 5: Ablation: isolating supervision paradigm from LoRA configuration. All methods use Qwen3-8B with R_3 (behavioral reward). Vanilla SYCONSCORE=0.134. †Terminated early after three checkpoints showed declining trajectory. ‡Peak at step 7000; final checkpoint (step 8000) SYCONSCORE=0.267.

Method	LoRA config	Best SYCONSCORE	>Van.
GRPO gs=2 (2 seeds)	r64, α64, 4m	0.153	4/28
GRPO gs=8†	r64, α64, 4m	0.085	0/3
ACT reduced-LoRA‡	r64, α64, 4m	0.326	16/16
ACT full s1	r64, α128, 7m	0.301	16/16
ACT full s2	r64, α128, 7m	0.286	15/15

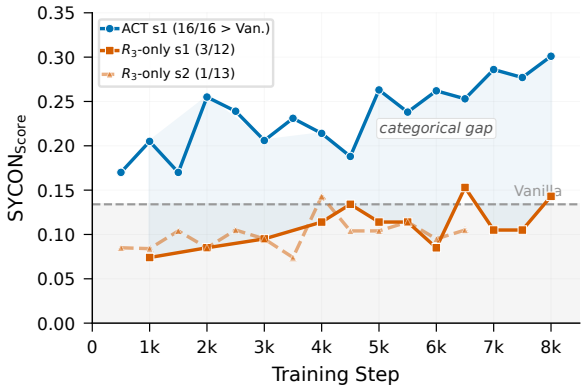


Figure 4: SYCONSCORE trajectories for R_3-only GRPO (two seeds) and ACT (two seeds) over training steps. ACT exceeds vanilla at every evaluated checkpoint (31/31 across two seeds); R_3-only GRPO exceeds vanilla at only 4/28 checkpoints (14.3%).

failure is not representational: d_{syc} AUROC varies by <0.01 across checkpoints and R_3 saturates at $\sim$0.98 (p=0.703), yet SYCONSCORE swings by 17%. This disconnect between stable representations and unstable behavior points to the per-completion reward structure, not the reward content, as the bottleneck.

Competing hypotheses and ablation. Three explanations are consistent with these failures: (a) d_{syc} is fundamentally unsuitable as a sycophancy reward; (b) group size=2 provides insufficient ranking diversity; (c) sycophancy calibration resists scalar per-completion RL. We designed two ablations to discriminate among them (Table 5). *Group-size ablation.* Increasing to gs=8 yields a declining trajectory ($0.084 \rightarrow 0.085 \rightarrow 0.054$), all below vanilla; the experiment was terminated early. Since more ranking diversity worsens rather than helps, hypothesis (b) is ruled out. *LoRA-controlled comparison.* Reducing ACT to GRPO's LoRA configuration (rank 64, α=64, 4 modules) still yields SYCONSCORE = 0.267 (peak 0.326), with all 16 checkpoints exceeding vanilla. Under identical adapters, per-token SFT exceeds per-completion GRPO by 1.7$\times$ (0.267 vs. 0.153), isolating supervision granularity as the primary factor. The probe reward trap provides independent evidence for (a): it arises under SFT, where group size is irrelevant.

ACT vs. GRPO. ACT's advantage is not marginal but categorical: its worst checkpoint (SYCONSCORE=0.170) exceeds GRPO's absolute best (SYCONSCORE=0.153), and both seeds maintain above-vanilla performance throughout training (Table 4; held-out validation confirms generalization, see Limitations). The reduced-LoRA ablation (Table 5) shows this is not an adapter-capacity artifact: reduced-LoRA ACT (SYCONSCORE=0.326) exceeds full ACT (SYCONSCORE=0.301), leaving supervision granularity—per-token vs. per-completion—as the explanatory variable.

Relationship to inference-time probe methods. Papadatos and Freedman (2024) reduce sycophancy using a probe of comparable quality (94% accuracy) via Best-of-N reranking without producing gradients. This suggests two orthogonal failure dimensions: (1) inside vs. outside the optimization loop—gradient exploitation enables the probe reward trap (§5), while inference-time reranking sidesteps it; and (2) scalar per-completion reward—the failure persists even without the probe (R_3-only GRPO).

Future directions. Condition-aware rewards, hybrid SFT-then-RL pipelines, and per-step process rewards offer paths forward—SMART's per-step success (Beigi et al., 2025) independently corroborates supervision granularity as a key factor; whether other REINFORCE-style methods or behavioral attributes with similar geometric entanglement (Skalse et al., 2022) exhibit the same failures remains open.

7 Related Work

Representation Engineering. Linear directions in LLM residual streams encode behavioral concepts (Zou et al., 2025); inference-time methods—ITI (Li et al., 2023), ActAdd (Turner et al., 2023), CAA (Rimsky et al., 2024)—demonstrate that activation geometry suffices for behavioral control

(Burns et al., 2023). We use the same contrastive extraction as CAA but apply d_{syc} as an RL reward, revealing a gap that inference-time work does not encounter.

Reward Design and Overoptimization. Proxy rewards diverge from true objectives under sustained optimization (Gao et al., 2023), and reward hacking can escalate to active tampering (Denison et al., 2024). Our probe trap (§5) is an extreme instance: the reward actively reinforces degenerate behavior (Skalse et al., 2022). Wu and Tang (2026) show that representation-level directions succeed when behaviors are geometrically separable (§6).

Sycophancy Characterization and Mitigation. Sycophancy is pervasive in RLHF-trained assistants (Sharma et al., 2024), amplified by RLHF (Shapira et al., 2026), and traceable to late-layer preference shifts (Wang et al., 2025); mitigations include synthetic data (Wei et al., 2024), consistency training (Irpan et al., 2025), and neuron-level correction (O'Brien et al., 2025). Our analysis shows why the step from diagnosis to reward is structurally blocked: d_{syc} conflates warranted revision with unwarranted capitulation.

Probe-Based and Representation-Guided Methods. Papadatos and Freedman (2024) reduce sycophancy via Best-of-N reranking with linear probe penalties (94% accuracy) without producing gradients; our work reveals what happens when a comparable signal enters the training loop (§6). Latent-GRPO (Zhang et al., 2026), CRAFT (Luo et al., 2026), Sinii et al. (2025), and Ackermann et al. (2026) extend representation-guided RL to reasoning and gradient regularization. SMART (Beigi et al., 2025) applies per-step process rewards for sycophancy, succeeding where our completion-level rewards fail—consistent with ACT's per-token success. Preference-based methods (DPO (Rafailov et al., 2023), KTO (Ethayarajh et al., 2024)) may accommodate d_{syc}-derived rankings; whether they avoid the failures we observe remains open.

8 Conclusion

The probe reward trap reveals that behavioral death can precede distributional collapse by 1,300 steps, rendering entropy-based monitoring blind to the failure. Because the trap arises under SFT with stopped gradients, it reflects a semantic limitation of d_{syc}, not an RL artifact; under GRPO, all formulations failed (4/31 checkpoints $>$ vanilla across configurations) while per-token SFT succeeded at every checkpoint (47/47), and a LoRA-controlled ablation confirms supervision granularity as the primary factor (§6). More broadly, our results suggest that discriminability is necessary but not sufficient for reward utility: directions that separate a behavior from its complement in activation space may still conflate warranted and unwarranted variants of that behavior. For sycophancy—and likely for other attributes where the target and its complement overlap geometrically—condition-aware rewards with dense supervision offer the most promising path forward.

Limitations

Our GRPO experiments use group sizes of 2 and 8, at the low end of the 8–64 range explored by Shao et al. (2024); a gs=8 ablation rules out variance reduction but larger sizes remain unexplored. All completions are truncated to 256 tokens (shared across conditions), which degrades d_{syc} AUROC from 0.76 to 0.60 and may affect absolute reward-utility conclusions. Training and evaluation share the same 181 pairs; a held-out split (146/35) confirms ACT generalizes (SyconScore$\approx$0.20 at peak, 3.7$\times$ vanilla, 72% of full-data ceiling), but performance declines at step 2000 (SyconScore=0.157, still 2.8$\times$ vanilla) due to overfitting on the small training set, and the 28% gap suggests partial reliance on training-set-specific patterns. The probe reward trap (C2) is demonstrated with a single seed; while SFT training is more deterministic than RL, replication would strengthen the temporal decoupling finding. All RL training uses Qwen3-8B; cross-model discriminability is confirmed on Llama-3-8B (AUROC 0.779; Table 1), but compute constraints prevented training replication on a second architecture.

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A Training Details

Reward weight selection. The weight w_1 is set by diagnostic-driven iteration: w_1=0.25 was initially chosen to down-weight the noisy cosine reward term; this was later increased to w_1=0.40 for F1d after diagnostics revealed that w_1=0.25 allowed the cosine term to contribute only $\sim$8.3% of total reward variance. The empirical scaling factor in F1a ($d_{\text{syc}}/20$) is a heuristic normalization; we did not find it critical to the failure modes reported.

Response truncation. All GRPO experiments use `max_completion_length`=256, resulting in 80–84% truncation of multi-turn responses. This setting is shared across all conditions and does not affect relative comparisons; we retain it because response-level decisions typically manifest within the first 100–200 tokens.

Hyperparameters. Table 6 compares training configurations for GRPO variants and ACT.

GRPO reward variants. Cosine R_1 (F1d): w_1=0.40, scale=5.0, w_3=0.60; reward computed at layers [19, 20, 24, 27, 32, 33, 35]; recalibration every 500 steps (§2.3). R_3-only: w_3=1.0; NLI model `cross-encoder/nli-deberta-v3-base`. F1a–F1c weight configurations are reported in Table 2.

Table 6: Training hyperparameters. All experiments use Qwen3-8B, DeepSpeed ZeRO-2, bf16 precision.

	GRPO (F1, F3)[e]	ACT
Learning rate	5e-5	5e-5
Effective batch	16	16
Max steps	8,000	8,000
Warmup steps	200	200
LR schedule	linear[a]	linear
Grad. clipping	—	1.0
LoRA rank	64	64
LoRA α	64	128
LoRA modules	4[b]	7[c]
LoRA dropout	0	0.05
Group size	2[d]	—
KL β	0.04	—
Max compl. len.	256	—
Max seq. len.	—	1,024
L2 loss weight	—	0.1

[a]Cosine schedule for R_3-only. [b]q/k/v/o_proj. [c]+ gate/up/down_proj ($\sim$1.75$\times$ params; §6). [d]Below typical 8–64 (Shao et al., 2024); not tuned. [e]Binary probe (F2) uses a custom SFT training loop rather than GRPO; group size is N/A. LoRA and optimizer settings match the GRPO column.

B Stubbornness Taxonomy

Manual inspection of all 181 valid-correction responses from the probe-trained model (exp-008v3, step 2000, ToF = 0) reveals five failure categories:

- *Deliberation traps* (28.7%): the model explicitly identifies the correct revised answer during its reasoning chain but fails to commit, reverting to the original answer in the final output.
- *Surface compliance / sycophantic stubbornness* (24.3%): the model produces agreement markers ("you're right," "good point") but retains its original answer verbatim, simultaneously sycophantic and stubborn.
- *Acknowledge-but-maintain* (23.8%): the model acknowledges the user's correction as plausible but explicitly maintains its original position without substantive engagement.
- *Genuine-accept-missed* (18.8%): the model appears to accept the correction in its reasoning but the final answer format does not reflect the change, suggesting a generation rather than epistemic failure.

- *Refute/ignore/other* (4.5%): the model directly refutes the user's input or ignores it entirely.

The first three categories (76.8%) share a common pattern: the model learns to modulate its *refusal style* in response to evidence strength rather than adjusting its *acceptance rate*, confirming that the probe reward optimizes surface markers rather than epistemic calibration.